E-commerce & Finance Insights Report

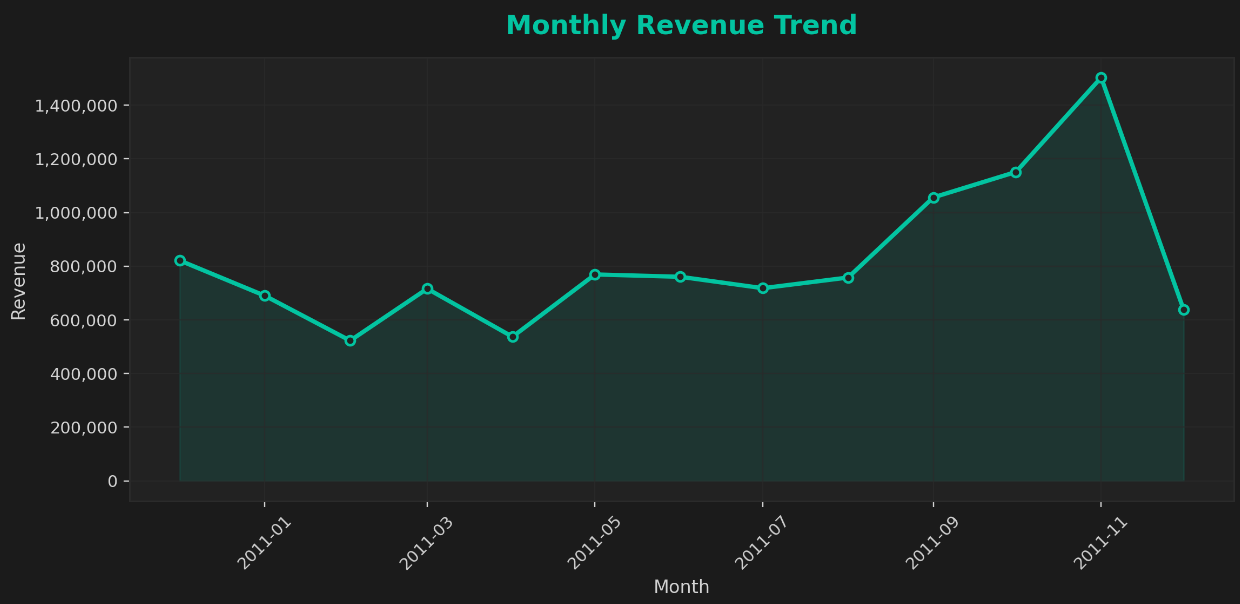
Generated via Python ReportLab • Author: Huzeif Khan

Key Performance Indicators (KPIs)

- **Total Revenue**: 10,641,558.95
- **Total Customers**: 4,338
- **Average Order Value**: 2,453.10

Visual Summary

Monthly Revenue Trend



Top 10 Products by Revenue

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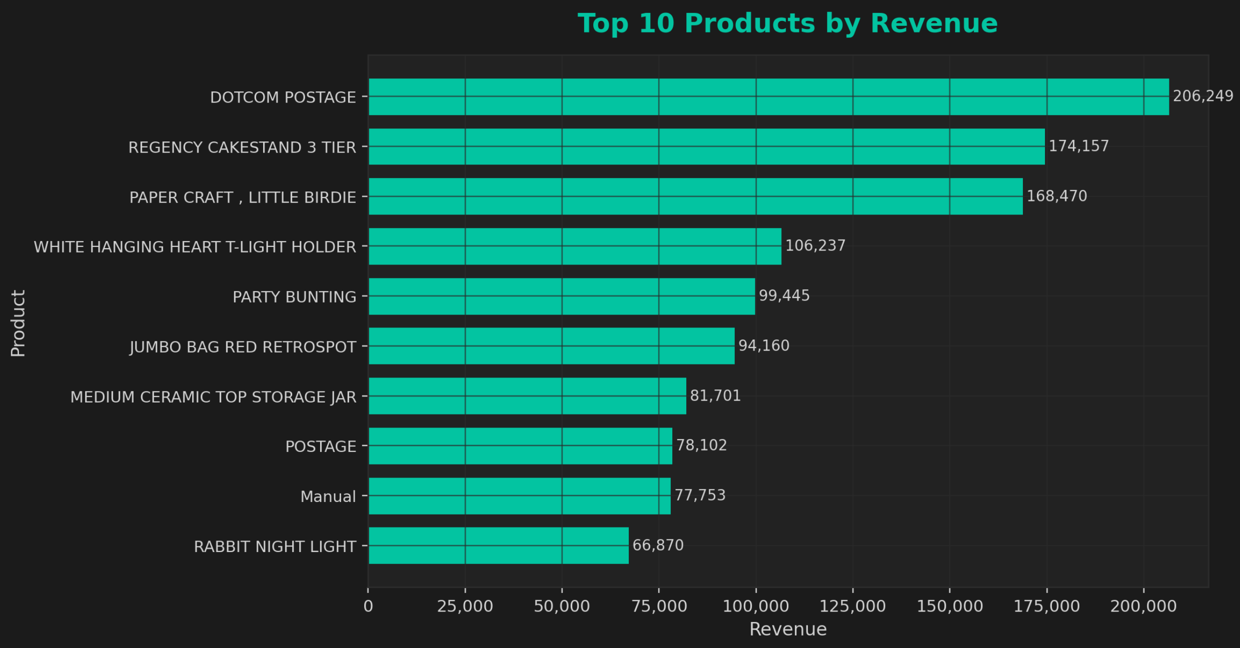
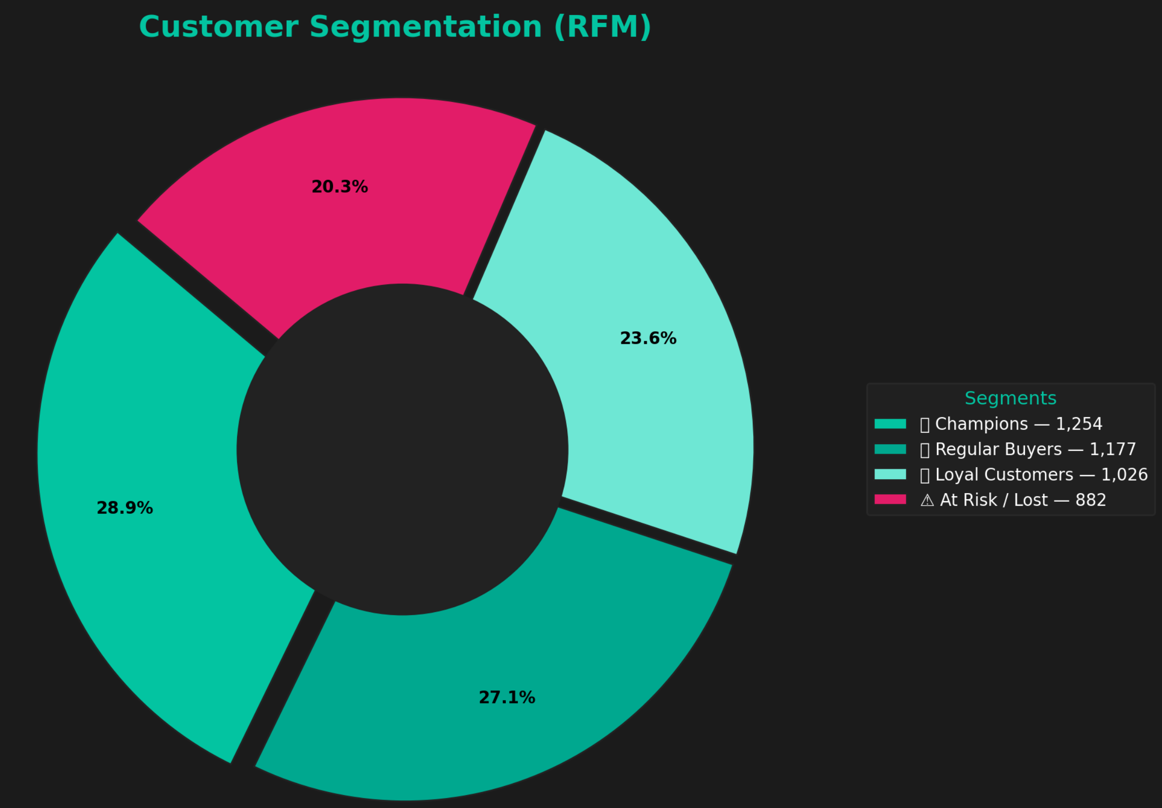
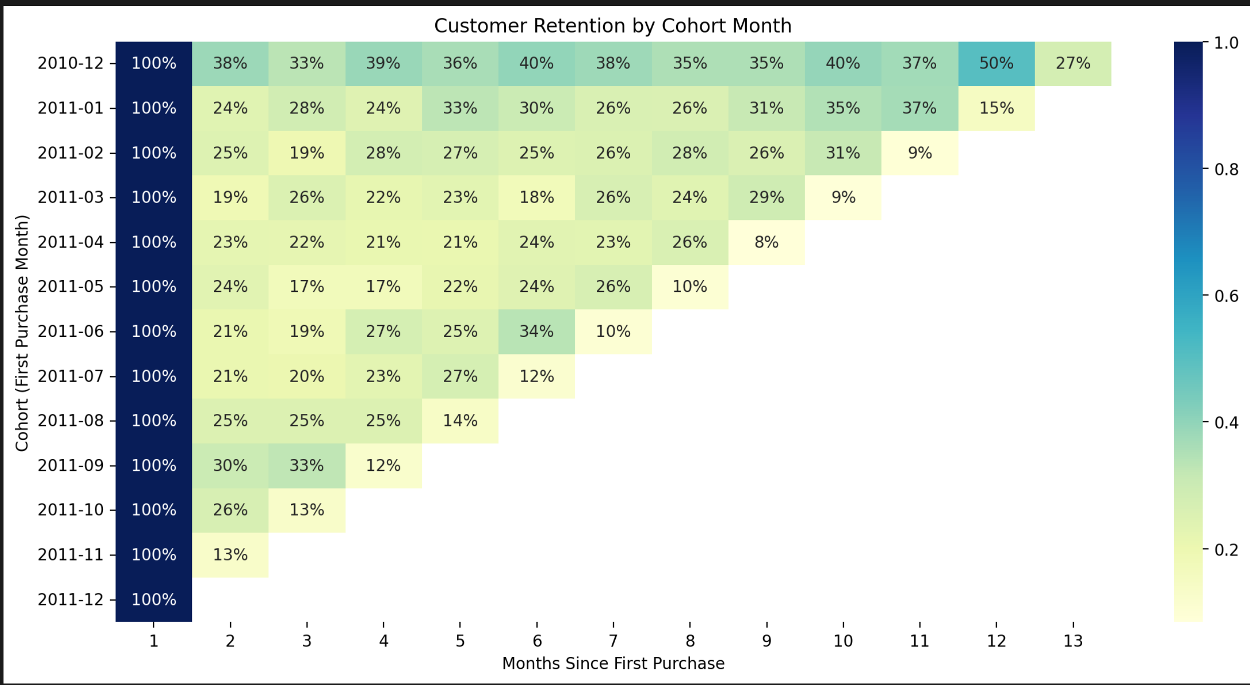


Customer Segmentation (RFM Model)



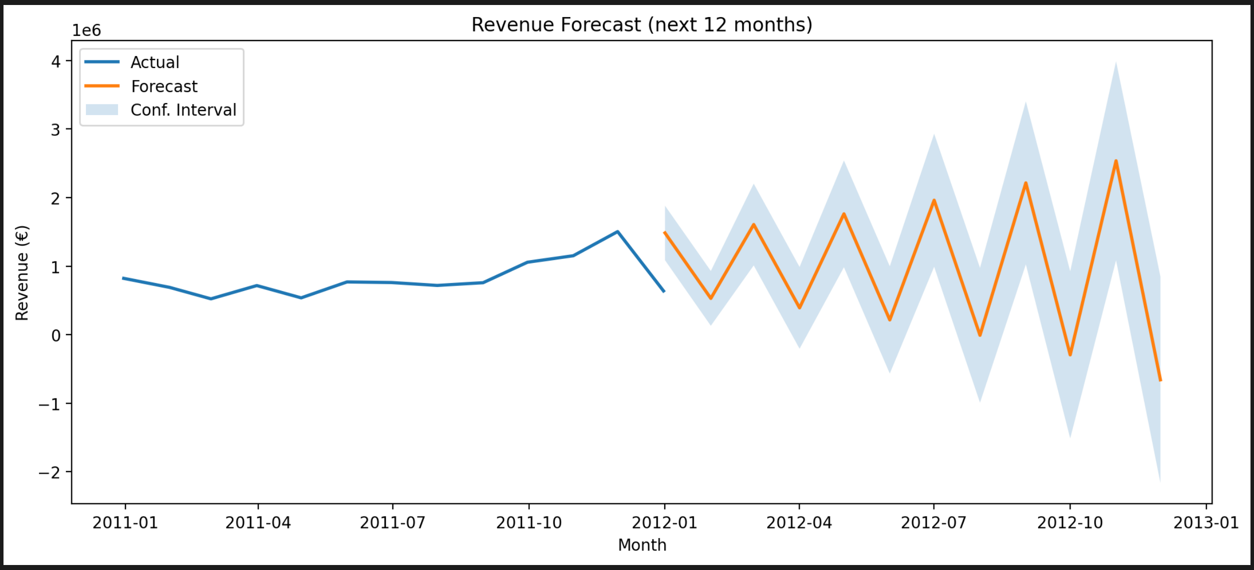
Cohort Retention

Each cell shows the % of customers who returned after N months, grouped by their first purchase month (cohort).



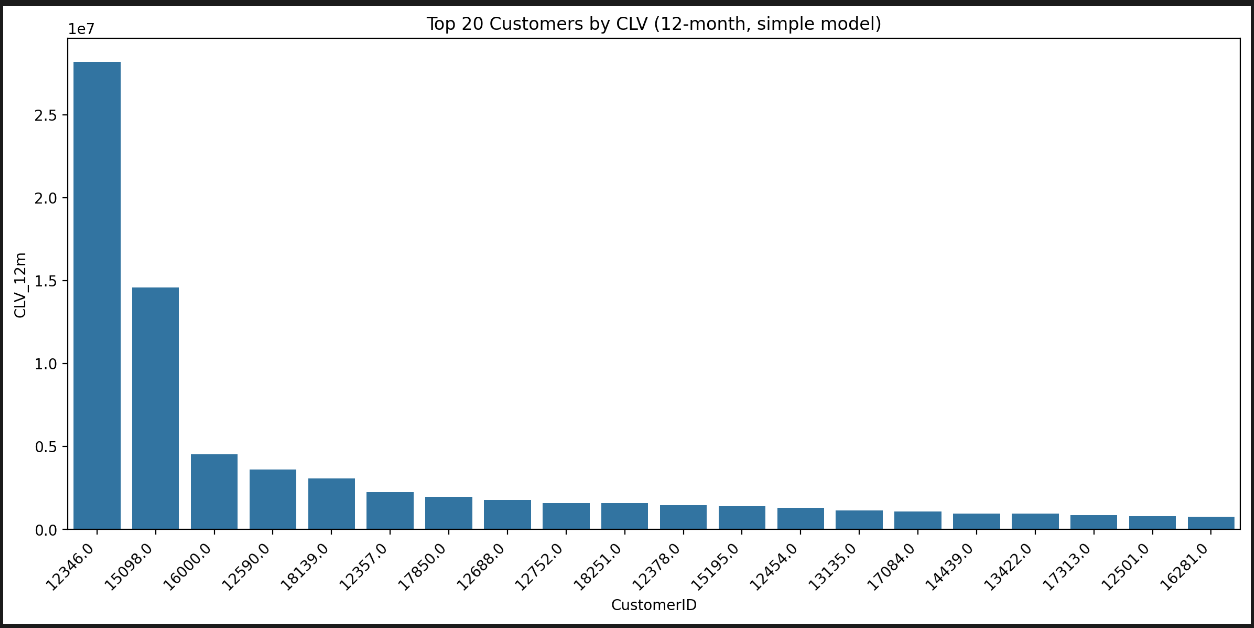
Revenue Forecast (Next 12 Months)

SARIMAX forecast with 80% confidence band. Use this to plan inventory, marketing budgets, and hiring.



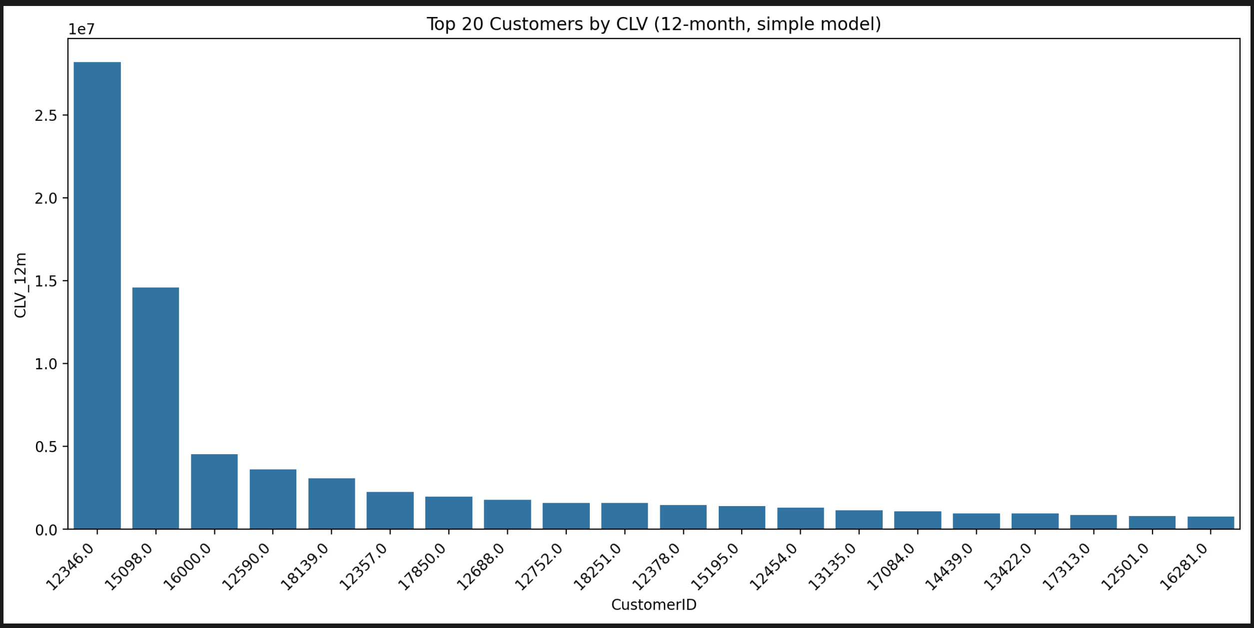
Customer Lifetime Value (CLV) - 6-Month Outlook

We estimate a simple 6-month CLV as AOV x (avg monthly repeat probability) x 6, added to the historical revenue. Top 20 customers shown.



Customer Lifetime Value (CLV v1 - 12-Month Model)

Deterministic CLV v1: AOV x Purchase Frequency per Month x 12 Months. Shows your top-value customers based on average order value and repeat purchase rate.



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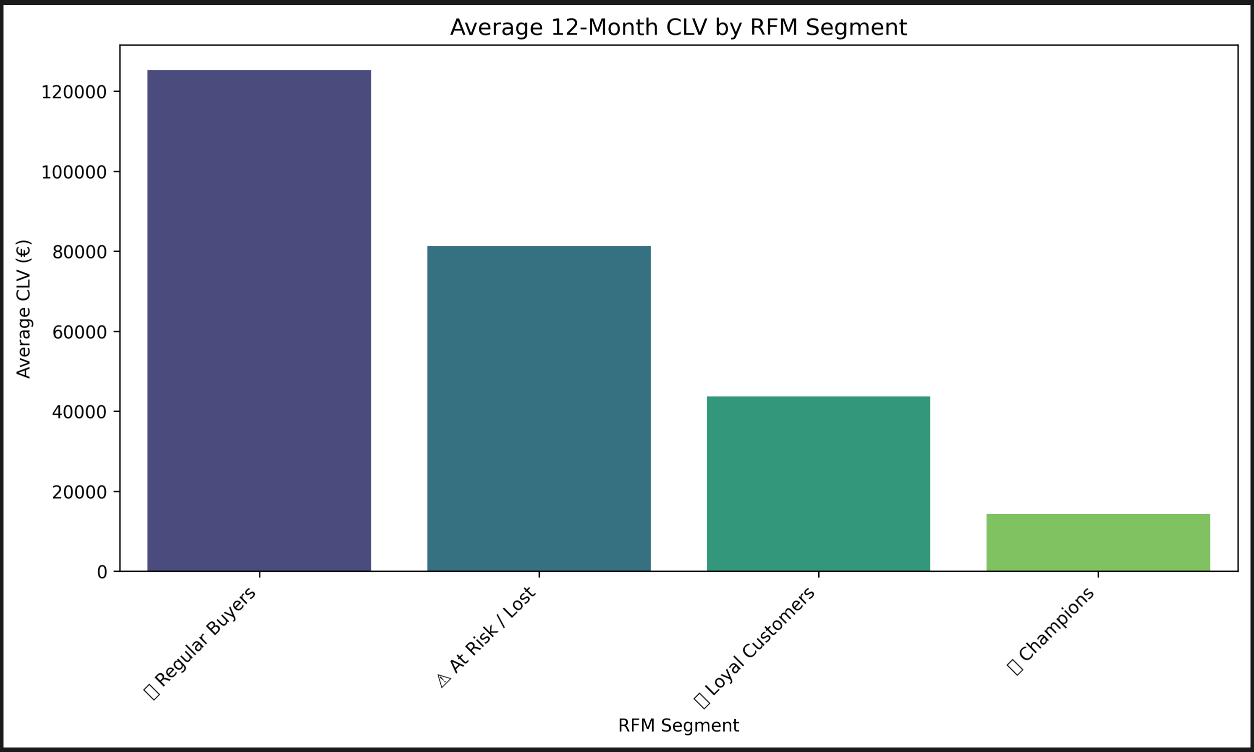
CLV by Customer Segment

Average predicted 12-month CLV for each RFM segment.

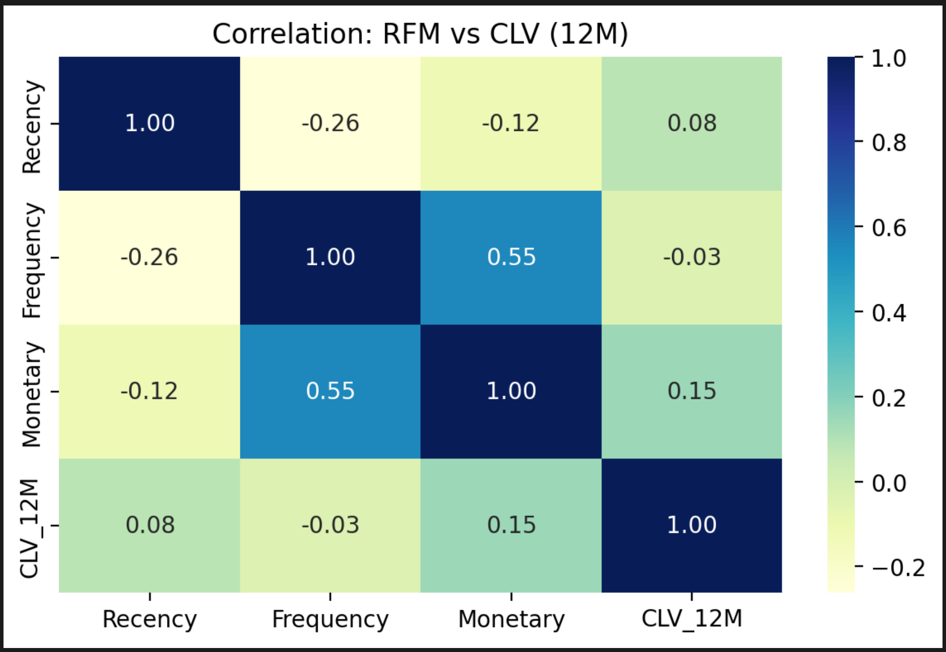


RFM x CLV - Segment Insights

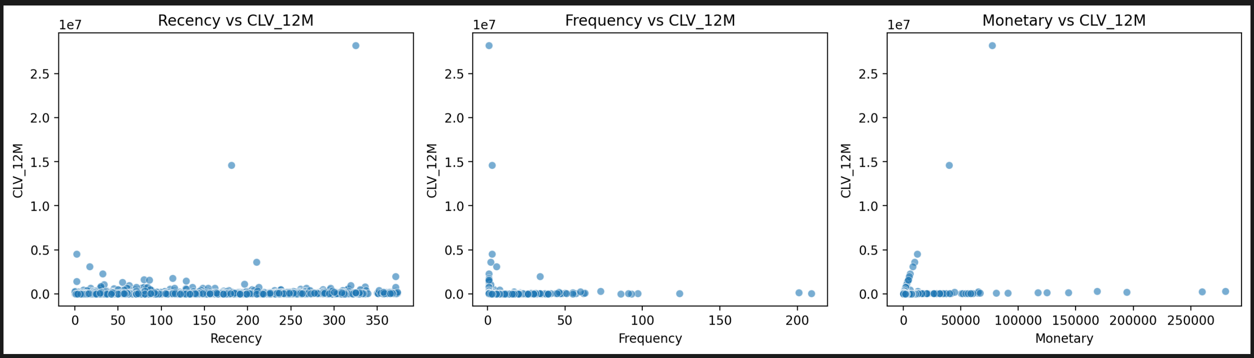
How lifetime value varies by customer segment, and how it correlates with Recency, Frequency and Monetary (RFM).

- ■ **Regular Buyers** - avg CLV â,¬125,288 (n=1177)
- ■■ **At Risk / Lost** - avg CLV â,¬81,282 (n=882)
- ■ **Loyal Customers** - avg CLV â,¬43,711 (n=1026)
- ■ **Champions** - avg CLV â,¬14,341 (n=1253)

RFM-CLV Correlation



CLV vs RFM (example scatter)

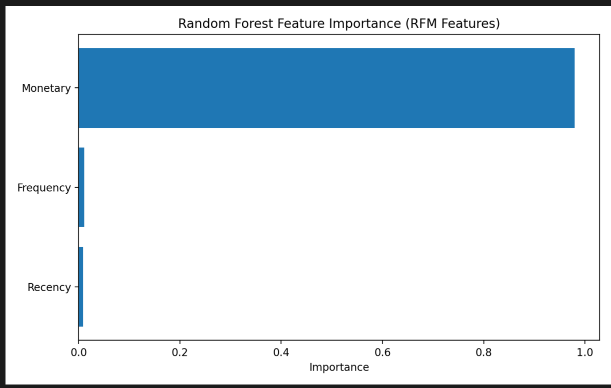


Predictive CLV Modelling (Machine Learning)

Model Performance (RFM-Based CLV Prediction)

Linear Regression
MAE: 0.00
RMSE: 0.00
R^2: 1.00

Random Forest Regressor
MAE: 43.66
RMSE: 1052.86
R^2: 0.987



Key Insights

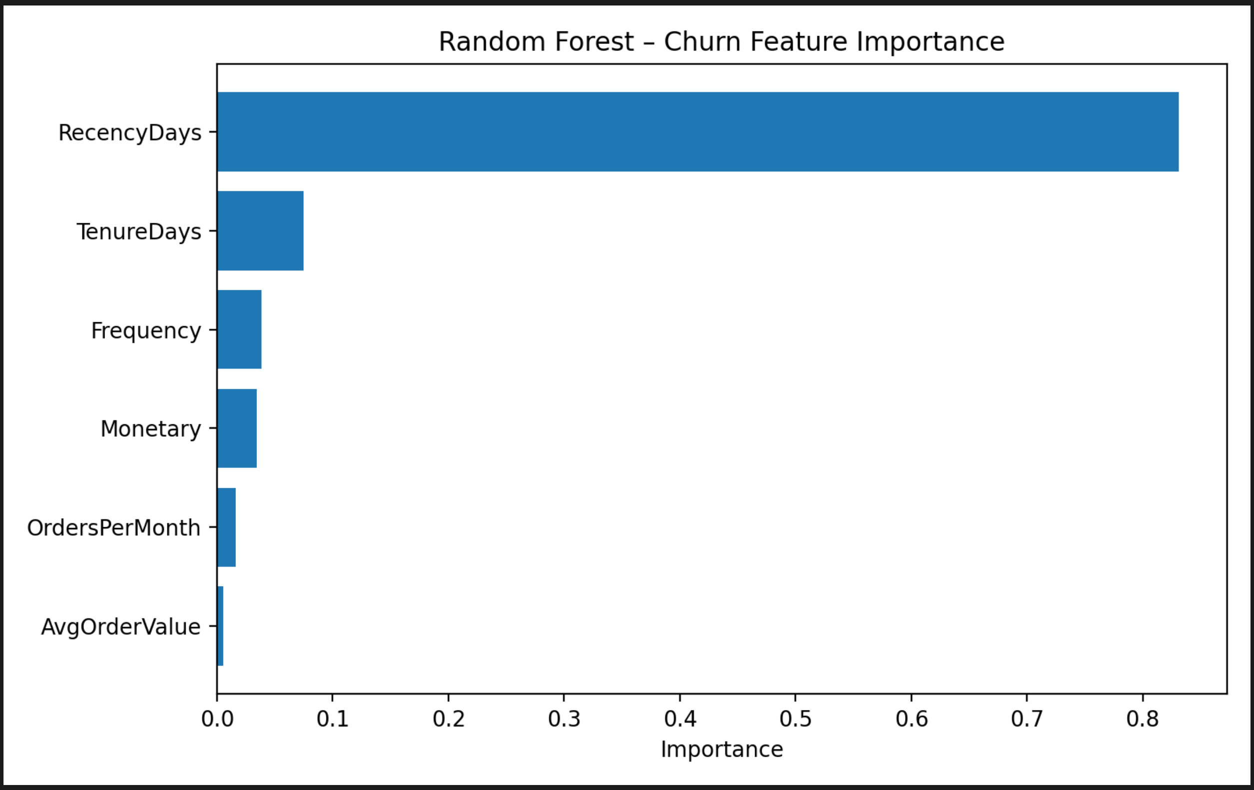
• RFM features (Recency, Frequency, Monetary) are exceptionally strong predictors of Customer Lifetime Value.
• Linear Regression achieved a perfect $R^2 = 1.00$ due to the deterministic relationship between Monetary value and CLV.
• Random Forest ($R^2 = 0.987$) confirms model stability and captures non-linear customer behaviours.
• Recency and Monetary value are the most influential features, reinforcing the validity of the RFM framework.
• Predictive modelling enhances segmentation accuracy and supports proactive customer value strategies.

Customer Churn Prediction (Phase 10)

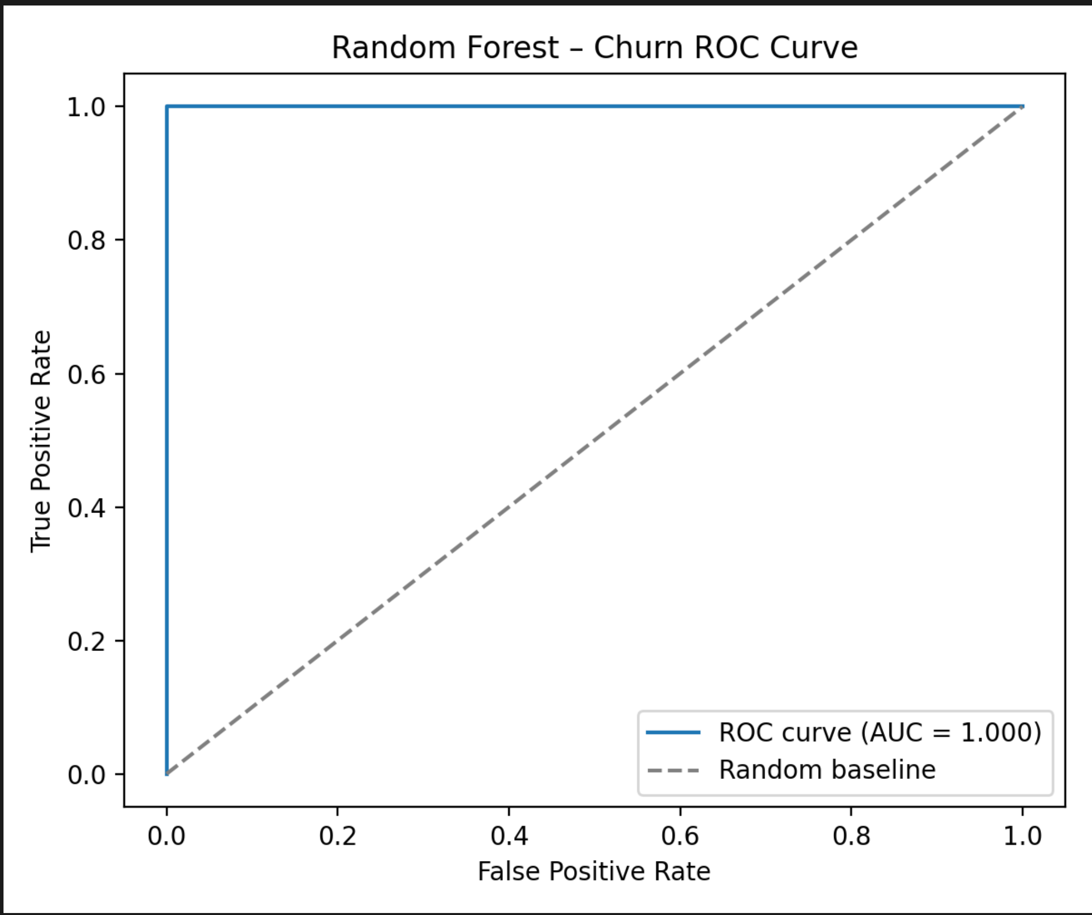
This section introduces a binary classification model predicting whether a customer is likely to become inactive based on purchase behavior. Churn is defined as no purchases in the last 90 days.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
LogisticRegression	0.992	1.000	0.975	0.987	1.000
RandomForestClassifier	1.000	1.000	1.000	1.000	1.000

Random Forest Feature Importance



Random Forest ROC Curve


Random Forest – Churn ROC Curve
True Positive Rate
False Positive Rate
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
ROC curve (AUC = 1.000)
Random baseline